

STATISTICS AND DECISION THEORY

Course	B.Tech.-III-Sem.	L	T	P	C
Course Code	25MABS31	3	-	-	3

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
CO1	apply sampling and estimation techniques for data analysis	4,9
CO2	perform hypothesis testing for statistical inference	4,9
CO3	analyze stochastic processes and Markov chains	4,9
CO4	apply decision theory methods for decision making	4,8,9
CO5	apply statistical learning concepts in AI/ML	4,9

Syllabus

Unit	Title/Topics	Hours
I	Sampling Distributions and Estimation	8
Random sampling and sampling distributions, Sampling distribution of mean, Central Limit Theorem, Point estimation (mean and variance), Interval estimation: confidence intervals for mean (known and unknown variance), Introduction to Bayesian estimation (basic idea only), Role of estimation in data-driven systems. Task: Compute expectation and variance from a sample dataset using Python.		
II	Hypothesis Testing and Statistical Inference	8
Hypothesis testing framework, Null and alternative hypotheses, Type-I and Type-II errors, Critical region and level of significance, Tests for single mean and difference of means, Hypothesis testing for proportions, Small sample tests: t-test, F-test, Statistical significance in model evaluation. Task: Perform hypothesis testing on two datasets using Python.		
III	Stochastic Processes and Markov Chains	4+4=8
Part A: Stochastic Processes: Introduction and classification of stochastic processes, Markov process (basic concept), Applications in computing systems (prediction, queueing – overview). Task: Simulate a simple stochastic process using Python.		
Part B: Markov Chains: Markov chains and transition probability matrix, Classification of states (basic idea), Steady-state probabilities, Applications: PageRank, recommendation systems. Task: Simulate a Markov chain and analyze steady-state behavior.		
IV	Decision Theory	8
Introduction to decision making and its importance. Elements of decision problems: states of nature, actions, payoffs. Decision tables and payoff matrices. Decision making under uncertainty: Maximax, Maximin, Minimax regret, Laplace and Hurwicz criteria. Decision making under risk: Expected Monetary Value (EMV), Expected Opportunity Loss (EOL). Applications in engineering and computing systems. Task: Construct a decision table and evaluate alternatives using decision criteria.		
V	Statistical Learning and Applications	8
Introduction to statistical learning, Regression (linear and logistic – basic concepts). Bias– variance tradeoff. Overfitting and underfitting, Model validation (cross-validation –		

basic idea). Applications in AI, machine learning, and data science.

Task: Implement linear regression model and evaluate using cross-validation.

Textbooks

1. Fundamentals of Mathematical Statistics - S. C. Gupta, V. K. Kapoor, S. Chand, 2020.
2. Introduction to Probability Models - Sheldon M. Ross, Academic Press, 2014.
3. Operations Research - S. D. Sharma, Kedar Nath Ram Nath & Co., Meerut, 2014.

CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
CO1	3	3	2	2	3	0	0	0	0	0	1	2	3
CO2	3	3	2	3	3	0	0	0	0	0	1	2	3
CO3	3	3	2	3	3	0	0	0	0	0	1	2	3
CO4	3	3	3	2	2	1	1	0	0	1	1	2	3
CO5	3	3	3	3	3	0	0	0	0	0	2	3	3

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
CO1	Applying sampling and estimation techniques supports engineering mathematics knowledge (P01), data-based problem analysis (P02), solution development (P03), and investigation of sample data (P04). Python-based computation supports modern tool usage (P05).	Supports statistical computation in software systems (PS01) and application of data analysis techniques to real-world problems (PS02).
CO2	Performing hypothesis testing strongly supports problem analysis (P02), statistical inference (P01), and investigation of data-driven claims (P04). Python implementation of statistical tests supports modern tool usage (P05).	Supports implementation of statistical validation methods in software systems (PS01) and application of inference techniques in AI/ML and data science contexts (PS02).
CO3	Analyzing stochastic processes and Markov chains supports engineering knowledge (P01), complex problem analysis (P02), and investigation of probabilistic systems (P04). Simulation of Markov chains supports modern tool usage (P05).	Supports modelling of probabilistic computing systems (PS01) and application of advanced techniques such as PageRank and recommendation systems (PS02).
CO4	Applying decision theory supports problem analysis (P02), solution development (P03), and evaluation of	Supports decision-support logic in software systems (PS01) and application of analytical decision methods to real-world

	alternatives under risk and uncertainty. It also relates to ethical decision-making (PO7), societal impact (PO6), and project/resource decisions (PO10).	engineering and computing problems (PS02).
C05	Applying statistical learning concepts strongly supports engineering knowledge (PO1), problem analysis (PO2), solution development (PO3), and investigation of model performance (PO4). Model validation using Python supports modern tool usage (PO5) and lifelong learning (PO11).	Strongly supports development of AI/ML-based software systems (PS01) and application of cutting-edge statistical learning techniques to real-world problems (PS02).

DESIGN AND ANALYSIS OF ALGORITHMS

Course	B.Tech.-III-Sem.	L	T	P	C
Course Code	25CSPC31	3	-	-	3

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
CO1	illustrate algorithm basics and analyze time and space complexity	4,9
CO2	implement disjoint sets and divide-and-conquer algorithms effectively	4,9
CO3	apply greedy and dynamic programming methods to optimization problems	4,9
CO4	solve problems using backtracking and branch-and-bound techniques	4,9
CO5	explain NP-Hard and NP-Complete problem concepts and classification	4,9

Syllabus

Unit	Title/Topics	Hours
I	Introduction	8
Introduction: Algorithm, pseudo code for expressing algorithms, performance analysis-space complexity, time complexity. Asymptotic Notations: big-oh, omega, theta, little oh and little omega notations. Task: Write a Program to perform operation count for a given pseudo code.		
II	Disjoint Sets, Divide and Conquer	8
Disjoint Sets: Disjoint set operations, UNION and FIND algorithms, biconnected components and AND/OR graph. Divide and Conquer: General method, applications-Binary search, Quick sort, Merge sort,		

Strassen' s matrix multiplication. Task: Write a Binary Search Program for a given list of values recursively and non-recursively.		
III	Greedy method and Dynamic Programming	4+4=8
Part-A: Greedy method: General method, applications-Job sequencing with deadlines, knapsack problem, Minimum cost spanning trees, Single source shortest path problem. Task: Write a Program to implement knapsack problem using greedy method.		
Part-B: Dynamic Programming: General method, applications - multistage graphs, optimal binary search trees, 0/1 knapsack problem, all pairs shortest path problem, travelling salesperson problem. Task: Write a Program for finding shortest path for multistage graph using dynamic programming.		
IV	Backtracking	8
Backtracking: General method, applications-n-queen problem, sum of subsets problem, graph coloring, Hamiltonian cycles. Branch and Bound: General method, applications - Travelling salesperson problem, 0/1 knapsack problem. Task: Write a program to find the optimal profit of a Knapsack using Branch and Bound Technique.		
V	NP-Hard and NP-Complete problems	8
NP-Hard and NP-Complete problems: Basic concepts, non-deterministic algorithms, NP - Hard and NP Complete classes, Cook' s theorem. Task: Write backtracking program to color graph nodes ensuring no adjacent nodes share colors.		
Textbooks		
1. Fundamentals of Computer Algorithms, Ellis Horowitz, Satraj Sahni and Rajasekaran, Galgotia Publications Pvt. Ltd. 2. Introduction to Algorithms, 2 nd Edition, T.H.Cormen, C.E.Leiserson, R.L.Rivest, and C.Stein, PHI Pvt. Ltd., Pearson Education.		
References		
1. Data structures and Algorithm Analysis in C++, Allen Weiss, 2 nd Edition, Pearson education. Design and Analysis of algorithms, Aho, Ullman and Hopcroft, Pearson education.		

CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
CO1	3	3	2	2	2	0	0	0	0	0	1	3	2
CO2	3	3	3	2	3	0	0	0	0	0	1	3	2
CO3	3	3	3	3	3	0	0	0	0	1	1	3	3
CO4	3	3	3	3	3	0	0	0	0	1	1	3	3
CO5	3	3	2	3	2	0	0	0	0	0	2	3	2

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
CO1	Illustrating algorithm basics and analyzing time-space complexity	Strongly supports algorithmic thinking for software system development (PS01) and

	strongly supports engineering knowledge (PO1) and problem analysis (PO2). Complexity evaluation supports solution comparison (PO3), investigation of algorithm efficiency (PO4), and lifelong learning (PO11).	moderately supports computational problem-solving technologies (PS02).
CO2	Implementing disjoint sets and divide-and-conquer algorithms strongly supports problem analysis (PO2), solution development (PO3), and use of programming tools (PO5). Algorithm implementation also involves investigation of correctness and efficiency (PO4).	Strongly supports implementation of efficient algorithms in software systems (PS01) and application of computational techniques to practical problems (PS02).
CO3	Applying greedy and dynamic programming methods strongly supports engineering knowledge (PO1), problem analysis (PO2), solution design (PO3), and investigation of optimization problems (PO4). These methods support efficient resource-oriented decisions (PO10).	Strongly supports development of optimized software solutions (PS01) and application of advanced algorithmic techniques to real-world problems (PS02).
CO4	Solving problems using backtracking and branch-and-bound strongly supports problem analysis (PO2), solution development (PO3), and investigation of constrained search problems (PO4). Implementation tasks support modern tool usage (PO5).	Strongly supports design of computational solutions for complex problems (PS01) and application of advanced optimization/search techniques (PS02).
CO5	Explaining NP-Hard and NP-Complete concepts supports engineering knowledge (PO1), complexity-based problem analysis (PO2), and investigation of computational limits (PO4). It promotes lifelong learning in advanced computing theory (PO11).	Strongly supports understanding of algorithmic feasibility in software systems (PS01) and moderately supports advanced computational problem classification (PS02).

DATABASE MANAGEMENT SYSTEMS

Course	B.Tech.-III-Sem.	L	T	P	C
Course Code	25CSPC32	3	-	-	3

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
CO1	design simple databases using database architectures	4,9
CO2	apply relational model, algebra, and calculus for data queries	4,9
CO3	implement SQL queries and optimize database performance	4,9
CO4	analyze schema design using normalization techniques	4,9
CO5	evaluate transaction management and NoSQL databases	4,9

Syllabus

Unit	Title/Topics	Hours
I	Introduction to Database Systems	8
Introduction: Introduction and applications of DBMS, Purpose of database, database architecture and structure - abstraction levels, data independence, database languages, database users and DBA roles. Introduction to Database Design: Database design process, data models, ER diagrams - entities, attributes, relationships, constraints, keys, generalization, specialization, aggregation, conceptual design with the E-R model for large enterprise.		
II	Relational Model, Algebra and Calculus	8
The Relational Model: Relational model concepts: schema, tuples, attributes. Integrity constraints (domain, key, referential). Querying relational data. Logical database design: ER to relational mapping, introduction to views. Creation, altering and dropping (destroying) of tables and views. Relational Algebra and Calculus: Relational algebra operators: Selection, projection, union, intersection, difference, joins. Relational calculus: tuple and domain relational calculus.		
III	SQL	4+4=8
Part-A: Basics of SQL, DDL, DML, DCL, structure – creation, alteration, defining constraints – Primary key, foreign key, unique, not null, check, in operator, Functions - aggregate functions, built-in functions – numeric, date, string functions, set operations.		
Part-B: Sub-queries, correlated sub-queries, Use of group by, having, order by, join and its types, exist, any, all, view and its types. Transaction control commands – commit, rollback, save point, cursors, stored procedures, Triggers. Indexing (B+ tree concept, index usage). Query optimization basics (execution plans, performance tuning).		
IV	Schema Refinement and Normal Forms	8
Schema Refinement and Normal Forms: Introduction to schema refinement, functional dependencies, reasoning about FDs. Normalization, normal forms: 1NF, 2NF, 3NF, BCNF. 4NF & 5NF – overview only. Decomposition: Lossless join, Dependency preservation.		
V	Transactions Management and NoSQL Databases	8
Transactions Management: Transaction concept and ACID properties, transaction state, Concurrent execution of transactions, Serializability and testing for Serializability, recoverability. Concurrency Control and Recovery System: Concurrency control, lock based protocols, time-stamp based protocols, validation based protocols, deadlocks (basic idea),		

crash recovery and backup system. Introduction to NoSQL databases: MongoDB, key-value, document, graph models. Use cases of NoSQL vs RDBMS (comparison).
Textbooks
1. Database Management Systems by Raghu Ramakrishnan and Johannes Gehrke, TMH, 2003.
2. Database System Concepts by Abraham Silberschatz, Henry F. Korth and S. Sudarshan, TMH, 2005.
References
1. Fundamentals of Database Systems, Elmasri Navate, Pearson Education, India.
2. Oracle SQL, P. K. Das Guptha and P Radha Krishna PHI.

CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
CO1	3	2	3	2	2	0	0	0	0	0	1	3	2
CO2	3	3	2	2	2	0	0	0	0	0	1	3	2
CO3	3	3	3	3	3	0	0	0	0	1	1	3	3
CO4	3	3	3	3	2	0	0	0	0	0	1	3	2
CO5	3	3	2	3	3	0	1	0	0	1	2	3	3

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
CO1	Designing simple databases using architectures strongly supports engineering knowledge (P01) and solution development (P03). ER modelling and database design involve problem analysis (P02), investigation of requirements (P04), and use of database design tools (P05).	Strongly supports database design as part of software system development (PS01) and application of data management technologies (PS02).
CO2	Applying relational model, algebra, and calculus strongly supports engineering knowledge (P01), problem analysis (P02), and formal query formulation (P03). Relational operations involve investigation of data retrieval logic (P04).	Strongly supports relational data modelling in software systems (PS01) and application of database query techniques in real-world systems (PS02).
CO3	Implementing SQL queries and optimizing performance strongly supports problem analysis (P02), solution development (P03), investigation of query execution (P04), and modern tool usage (P05). Performance tuning also supports project efficiency awareness (P010).	Strongly supports implementation of database-driven software systems (PS01) and application of modern DBMS technologies for efficient real-world solutions (PS02).
CO4	Analyzing schema design using	Strongly supports development of reliable

	normalization strongly supports problem analysis (PO2), solution development (PO3), and investigation of dependency and redundancy issues (PO4). It strengthens engineering knowledge of database design principles (PO1).	and well-structured database systems (PS01) and application of systematic design techniques in software solutions (PS02).
CO5	Evaluating transaction management and NoSQL databases supports engineering knowledge (PO1), problem analysis (PO2), and investigation of concurrency, recovery, and distributed data models (PO4). NoSQL tools support modern tool usage (PO5), ethics in data handling (PO7), project reliability (PO10), and lifelong learning (PO11).	Strongly supports development of reliable database-backed software systems (PS01) and application of advanced database technologies such as MongoDB and NoSQL models (PS02).

OOP THROUGH JAVA

Course	B.Tech.-III-Sem.	L	T	P	C
Course Code	25CSPC33	3	-	-	3

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
CO1	explain Java fundamentals and OOP concepts	4,9
CO2	apply inheritance, polymorphism, and interfaces	4,9
CO3	implement exception handling and multithreading	4,9
CO4	develop basic GUI applications and REST concepts	4,9
CO5	build database and web applications using Java	4,9

Syllabus

Unit	Title/Topics	Hours
I	Java Basics	8
Java Basics: History, installation, buzzwords. Data types, variables, scope, arrays. Operators, expressions, control statements. Type conversion and casting. Classes, objects, methods, constructors. Access control, this keyword. Garbage collection, parameter passing, recursion. String class. Introduction to Java Collections (ArrayList, HashMap – basics).		
II	Inheritance, Polymorphism, Packages & Interfaces	8
Inheritance: Types, super, final, Object class. Polymorphism: Overloading, overriding, dynamic binding. Abstract classes and methods. Packages: Define, create, import.		

Interfaces: Define, implement, extend. Interface variables. Functional interfaces and lambda expressions (basic introduction).		
III	Exception Handling & Multithreading	4+4=8
Part-A: Exception Handling: Concepts, hierarchy, try, catch, throw, throws, finally. Built-in exceptions. User-defined exceptions.		
Part-B: Multithreading: Thread life cycle, Creating threads, Thread priorities, Synchronization, Inter-thread communication. Executor framework basics (thread pools concept).		
IV	Event Handling & GUI / Application Layer	8
Event handling: Event model, listeners, adapters. Mouse & keyboard events. Swing: Components (JFrame, JLabel, JButton, etc.), Layout managers (Flow, Border, Grid). Focus on basic GUI only (intro level). Introduction to REST API concepts (request/response, JSON basics).		
V	Servlets with JDBC	8
Servlets: Lifecycle, architecture, Handling client requests, HTTP methods, status codes, Session tracking (cookies). JDBC: Architecture, Drivers, statements, CRUD operations. Introduction to RESTful services using Java (conceptual). Basic API design and JSON data handling.		
Textbooks		
1. Herbert Schildt - Java: The Complete Reference (11th Edition), MGH, 2018. 2. Core Java Volume I – Fundamentals - Cay S. Horstmann, Pearson, 2021. 3. Joshua Bloch - Effective Java (advanced best practices), Addison-Wesley, 2018.		

CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	3	2	2	1	3	0	0	0	0	0	1	3	2
C02	3	3	3	2	3	0	0	0	0	0	1	3	2
C03	3	3	3	2	3	0	0	0	0	0	1	3	2
C04	3	2	3	2	3	0	0	1	1	0	1	3	3
C05	3	3	3	3	3	0	0	1	1	1	1	3	3

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
C01	Explaining Java fundamentals and OOP	Strongly supports foundational software

	concepts supports engineering knowledge (P01), problem analysis (P02), and basic solution development (P03). Programming practice supports modern tool usage (P05) and lifelong learning (P011).	development using OOP principles (PS01) and application of programming technologies (PS02).
C02	Applying inheritance, polymorphism, and interfaces strongly supports problem analysis (P02), solution design (P03), and investigation of object-oriented relationships (P04). Java implementation supports tool usage (P05).	Strongly supports modular and reusable software system design (PS01) and application of object-oriented technologies (PS02).
C03	Implementing exception handling and multithreading supports engineering knowledge (P01), problem analysis (P02), and solution development (P03). Thread synchronization and error handling involve investigation (P04) and tool usage (P05).	Strongly supports development of robust and concurrent software systems (PS01) and application of modern programming techniques (PS02).
C04	Developing GUI applications and understanding REST concepts supports solution development (P03), problem analysis (P02), and modern tool usage (P05). Event handling and UI design support teamwork (P08) and communication (P09).	Strongly supports application development using Java frameworks (PS01) and application of modern technologies such as REST APIs and UI systems (PS02).
C05	Building database and web applications using Java strongly supports engineering knowledge (P01), problem analysis (P02), solution development (P03), and investigation of application workflows (P04). Use of JDBC and servlets strongly supports tool usage (P05) and project awareness (P010).	Strongly supports development of full-stack software systems (PS01) and application of modern web and database technologies (PS02).

Course	B.Tech.-III-Sem.	L	T	P	C
Course Code	25CSPC34	3	-	-	3

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
CO1	apply software engineering process models and requirements engineering	4,9
CO2	design software systems using architectural and component-level design	4,9
CO3	model software systems using UML diagrams	4,9
CO4	apply testing strategies and quality assurance techniques	4,9
CO5	analyze risks and design scalable systems using modern principles	4,9,12

Syllabus

Unit	Title/Topics	Hours
I	Introduction and Requirements Engineering	8
Introduction to Software Engineering, evolving role of software, SDLC models: Waterfall, Incremental, Agile (overview), software engineering as a layered technology, Capability Maturity Model Integration (CMMI) – overview, functional and non-functional requirements, requirements engineering process: feasibility study, requirements elicitation & analysis, validation & management, software requirements specification (SRS). <i>Task: Document functional and non-functional requirements for online bookstore system.</i>		
II	Design Engineering & System Design (LLD + HLD)	8
Design process and design quality; design concepts: abstraction, modularity, cohesion, coupling; architectural design (HLD): software architecture, architectural styles (layered, client-server, microservices (overview)). Scalability concepts: load balancing, caching. component-level design (LLD): class design, API design basics. user interface design: golden rules, UI/UX basics. <i>Task: Design high-level architecture for an online bookstore using layered architecture.</i>		
III	UML Modeling and Design Visualization	4+4=8
Part-A: Structural Modeling: Principles of modeling, UML overview. Class, object, component diagrams (overview). <i>Task: Design a class diagram for ATM banking system / library management system</i>		
Part-B: Behavioral Modeling: Use case, Sequence, Activity, State chart, Deployment diagrams (overview). <i>Task: Draw use case diagram for ATM operations, sequence diagram for transaction flow.</i>		
IV	Testing & Quality Assurance	8
Software testing strategies, black-box testing & white-box testing, unit testing, integration testing, system testing, validation testing and debugging, software quality metrics, product and process metrics, introduction to automated testing (Selenium/PyTest). <i>Task: Write unit test cases for login module, transaction/payment module.</i>		
V	Software Management & System Design Case Studies	8
Risk management: risk identification, analysis, mitigation, RMMM plan, software quality assurance (SQA), software reviews and reliability, ISO 9000 quality standards (overview), basic system design case studies: URL shortener, e-commerce, chat system (overview). <i>Task: Identify and mitigate risks in e-commerce system design project.</i>		
Textbooks		
1. Roger S. Pressman, Software engineering - A practitioner's Approach, TMH (I), 8 th Edn,		

2015. 2. Ian Sommerville, Software Engineering, Pearson education Asia, 10 th Edn, 2015. 3. Grady Booch et al., The Unified Modeling Language User Guide, 2nd Ed., Pearson, 2005.
References
1. Designing Data-Intensive Applications - Martin Kleppmann, 2017, O’ Reilly Media. 2. Clean Architecture - Robert C. Martin, 2017, Prentice Hall (Pearson). 3. Head First Design Patterns - Eric Freeman and Elisabeth Robson.

CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PS01	PS02
C01	3	3	2	2	2	1	1	1	1	1	1	3	2
C02	3	3	3	2	2	1	0	1	1	1	1	3	3
C03	3	2	3	2	3	0	0	1	2	0	1	3	2
C04	3	3	2	3	3	0	1	1	1	1	1	3	2
C05	3	3	3	3	3	1	1	1	1	1	2	3	3

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
C01	Applying software process models and requirements engineering strongly supports engineering knowledge (PO1) and problem analysis (PO2). Requirements elicitation and SRS development support solution understanding (PO3), investigation (PO4), ethical considerations (PO7), teamwork (PO8), communication (PO9), and project planning (PO10).	Strongly supports structured SDLC-based software development (PS01) and moderate application of modern engineering practices (PS02).
C02	Designing software systems using architectural and component-level design strongly supports solution development (PO3), problem analysis (PO2), and engineering knowledge (PO1). Concepts like scalability and architecture relate to project planning (PO10) and real-world system impact (PO6).	Strongly supports system design and development using SDLC principles (PS01) and application of modern scalable architectures (PS02).
C03	Modeling systems using UML diagrams supports engineering knowledge (PO1), solution design (PO3), and strong use of modeling tools (PO5). UML diagrams enhance communication (PO9) and teamwork (PO8).	Strongly supports system modeling in software design (PS01) and application of modern visualization tools (PS02).
C04	Applying testing strategies and quality assurance strongly supports problem	Strongly supports development of reliable and tested software systems (PS01) and

	analysis (P02), investigation (P04), and tool usage (P05). Testing processes relate to ethical responsibility (P07), teamwork (P08), and project quality assurance (P010).	application of modern QA/testing tools (PS02).
C05	Analyzing risks and designing scalable systems strongly supports engineering knowledge (P01), problem analysis (P02), solution development (P03), and investigation of complex systems (P04). Risk management and scalability relate to societal impact (P06), ethics (P07), project management (P010), and lifelong learning (P011).	Strongly supports development of scalable and robust software systems (PS01) and application of advanced system design principles in real-world scenarios (PS02).

INNOVATION AND ENTREPRENEURSHIP

Course	B.Tech.-III-Sem.	L	T	P	C
Course Code	25MSHS31	2	-	-	2

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
C01	interpret innovation concepts and entrepreneurial roles	4,8,9
C02	identify customer problems and generate innovative solutions	4,8,9
C03	evaluate opportunities and develop basic prototypes	4,8,9
C04	apply basic business and financial models	4,8
C05	explain startup lifecycle and IPR concepts	4,8,9

Syllabus

Unit	Title/Topics	Hours
I	Fundamentals of Innovation and Entrepreneurship	6
Innovation: Importance, key features, types, applications in manufacturing and services, and planning for implementation. Entrepreneurship: Introduction, types, essential attributes, entrepreneurial mindset, role in economic development, women entrepreneurship, significance of on-campus startups, and building networks while on campus. Task: Case study on a successful startup.		
II	Problem and Customer Identification	6
Identification of gaps and problems, analyze industry context, segment markets and customers, validate and iterate problem-solution fit, map competitors and trends, generate ideas using design thinking and ideation techniques. Task: Problem-solution mapping exercise.		

III	Opportunity assessment and Prototype development	3+3=6
Part-A: Identify and map global competitors, review industry trends, and understand market sizing. Assessing scope and potential scale for the opportunity. Task: <i>Analyze competitors, trends, and estimate market size for opportunity.</i>		
Part-B: Understanding prototyping and Minimum Viable Product (MVP). Developing a prototype: Testing, and validation. Task: <i>Create simple prototype idea.</i>		
IV	Business & Financial Models	6
Introduction to business models and Lean Canvas (9-block) approach. Prepare financial plan, understand unit economics and economies of scale, analyze performance, and design Go-To-Market strategy including channels, digital presence, and customer acquisition. Task: <i>Prepare simple Lean Canvas.</i>		
V	Startups and IPR	6
Startups: Overview of startup requirements, team building, mentors, pitch preparation, registration, funding schemes, institutional support, lifecycle, documentation, legal aspects, and National Innovation Startup Policy (NISP). IPR: Patents, designs, copyrights, trademarks, geographical indications, licensing, and technology transfer. Task: <i>Prepare startup pitch outline.</i>		
Textbooks		
1. John R. Bessant, Joe Tidd, Innovation and Entrepreneurship, 4 th Edition, Wiley. 2. Ajay Batra, Startup Launch Book, 2020.		

CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
CO1	2	2	1	1	1	2	1	1	1	1	2	1	2
CO2	2	3	2	2	1	2	1	2	2	1	2	1	2
CO3	2	3	3	2	2	2	1	2	2	2	2	2	3
CO4	2	2	3	1	1	2	1	1	2	3	2	1	2
CO5	2	2	2	1	1	2	3	1	2	2	2	1	2

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
CO1	Interpreting innovation and entrepreneurial roles supports understanding of engineering concepts in societal context (P01, P06). It develops analytical thinking (P02), ethical awareness (P07), teamwork (P08), communication (P09), and	Provides foundational understanding for applying technology in entrepreneurial settings (PS02) and limited linkage to software/system design (PS01).

	promotes lifelong learning (P011).	
CO2	Identifying customer problems and generating solutions strongly supports problem analysis (P02), solution development (P03), and investigation of real-world needs (P04). It also enhances teamwork (P08), communication (P09), and societal awareness (P06).	Supports problem-driven system design thinking (PS01) and application of innovative solutions using modern technologies (PS02).
CO3	Evaluating opportunities and developing prototypes strongly supports solution design (P03), problem analysis (P02), and investigation (P04). It also involves teamwork (P08), communication (P09), project planning (P010), and understanding societal impact (P06).	Supports development of prototype-based systems (PS01) and strong application of emerging technologies for real-world solutions (PS02).
CO4	Applying business and financial models supports solution planning (P03), understanding economic aspects (P010), and societal awareness (P06). It also enhances communication (P09) and decision-making abilities (P02).	Provides limited support for software/system lifecycle awareness (PS01) and moderate application of technology in business contexts (PS02).
CO5	Explaining startup lifecycle and IPR concepts supports engineering ethics (P07), societal responsibility (P06), communication (P09), and project/legal awareness (P010). It also promotes continuous learning (P011).	Supports awareness of intellectual property in software/system development (PS01) and application of innovation and technology commercialization (PS02).

COMPUTATIONAL MATHEMATICS LAB

Course	B.Tech.-III-Sem.	L	T	P	C
Course Code	25MABS33	-	-	2	1

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
CO1	compute eigenvalues and solve nonlinear equations	4,9

C02	solve linear systems using iterative methods	4,9
C03	apply numerical methods for differential equations	4,9
C04	perform curve fitting using least squares	4,9
C05	analyze and visualize real datasets	4,9

List of Experiments

Note: 1. Visualize all solutions graphically through programmes using Python/MATLAB/SCI-Lab.
2. Faculty provides test cases for each experiment during labs for practice and validation.

Week	Title/Experiment
1	Find the real and complex Eigen values and Eigen vectors.
2	Find the root of a given equation using Bisection method.
3	Find the root of a given equation Newton Raphson Method.
4	Solve a system of linear equations using Jacobi method.
5	Solve a system of linear equations using Gauss-Seidel method.
6	Solve exact and non-exact differential equations.
7	Solve exponential growth/decay and Newton's law of cooling problems.
8	Solve homogeneous and non-homogeneous ODEs.
9	Solve second-order differential equations using Runge-Kutta method.
10	Fit a straight line and second-degree polynomial using least squares method.
11	Fit power and exponential curves using least squares method.
12	Perform data fitting and visualization on real datasets and interpret results.
References	
1. Computational Mathematics Lab Manual, Dept. of FED, CMRIT, Hyd.	
Micro-Projects: Student should submit a report on <i>design, implement, test and debug</i> from one of the following/any other micro-project(s) approved by the faculty before the deadline.	
1. Numerical Root Finding Visualizer using Python. 2. Linear System Solver Dashboard using Iterative Methods. 3. ODE Solver for Real-World Dynamic Systems. 4. Curve Fitting and Regression Analysis using Real Data. 5. Eigenvalue-Based Image Compression using PCA. 6. Stability Analysis of Engineering Systems using Eigen Methods. 7. Heat Transfer Simulation using Finite Difference Method. 8. Optimization Techniques for Resource Allocation Problems. 9. Traffic Flow Modeling using Differential Equations. 10. Data-Driven Forecasting using Mathematical Models.	

Test Cases

Week	Test Cases for Experiments			
1	Eigenvalues and Eigenvectors			
	Type	Input	Expected Output	Explanation
	Success	2× 2 real matrix	Real eigenvalues	Validates basic calculation
	Success	Matrix with complex roots	Complex eigenvalues	Checks complex handling
	Success	Symmetric matrix	Orthogonal eigenvectors	Tests correctness
	Failure/Edge	Non-square matrix	Error	Eigenvalues need square matrix
	Failure/Edge	Singular matrix	Zero eigenvalue	Edge case
2	Bisection Method			
	Type	Input	Expected Output	Explanation
	Success	Valid interval	Root approximated	Validates bisection
	Success	Root inside interval	Converges	Checks sign change
	Success	Small tolerance	Accurate root	Tests precision
	Failure/Edge	No sign change	Error	Invalid interval
	Failure/Edge	Same interval points	Error	Boundary case
3	Newton-Raphson Method			
	Type	Input	Expected Output	Explanation
	Success	Valid initial guess	Root found	Validates iteration
	Success	Polynomial equation	Correct root	Checks derivative use
	Success	Good tolerance	Accurate result	Tests convergence
	Failure/Edge	Derivative zero	Error	Avoids division by zero
	Failure/Edge	Poor initial guess	No convergence	Edge case
4	Jacobi Method			
	Type	Input	Expected Output	Explanation
	Success	Diagonally dominant system	Solution obtained	Validates Jacobi
	Success	Given tolerance	Iterations converge	Checks stopping condition
	Success	3-variable system	Correct approximation	Tests matrix handling
	Failure/Edge	Non-dominant matrix	May not converge	Edge case
	Failure/Edge	Zero diagonal element	Error	Division by zero

5	Gauss-Seidel Method			
	Type	Input	Expected Output	Explanation
	Success	Diagonally dominant system	Solution obtained	Validates method
	Success	Initial guess	Converges faster	Checks iteration update
	Success	3× 3 system	Correct approximation	Tests implementation
	Failure/Edge	Non-convergent system	No convergence	Edge case
	Failure/Edge	Zero diagonal element	Error	Invalid system
6	Exact and Non-exact Differential Equations			
	Type	Input	Expected Output	Explanation
	Success	Exact DE	Solution found	Validates exactness
	Success	Non-exact DE	Identified correctly	Checks classification
	Success	Integrating factor	Solved DE	Tests method selection
	Failure/Edge	Invalid DE form	Error	Input validation
	Failure/Edge	Missing variables	Cannot solve	Boundary case
7	Growth/Decay and Cooling Problems			
	Type	Input	Expected Output	Explanation
	Success	Growth rate positive	Population increases	Validates growth model
	Success	Decay rate positive	Quantity decreases	Checks decay model
	Success	Cooling data	Temperature predicted	Tests Newton cooling
	Failure/Edge	Negative time	Error	Invalid physical input
	Failure/Edge	Missing rate constant	Error	Required parameter
8	Homogeneous and Non-homogeneous ODEs			
	Type	Input	Expected Output	Explanation
	Success	Homogeneous ODE	Complementary function	Validates solution
	Success	Non-homogeneous ODE	Complete solution	Checks PI + CF
	Success	Initial conditions	Particular solution	Tests constants
	Failure/Edge	Invalid order	Error	Checks equation form
	Failure/Edge	Missing initial values	General solution only	Edge case
9	Second-order DE using Runge-Kutta			
	Type	Input	Expected Output	Explanation

	Success	Valid second-order ODE	Approximate solution	Validates RK method
	Success	Small step size	More accurate output	Checks numerical accuracy
	Success	Initial conditions	Correct iterations	Tests conversion to first-order
	Failure/Edge	Step size = 0	Error	Boundary case
	Failure/Edge	Missing initial condition	Error	Required for RK
10	Least Squares: Line and Polynomial			
	Type	Input	Expected Output	Explanation
	Success	Linear data	Best-fit line	Validates line fitting
	Success	Quadratic data	Polynomial fit	Checks degree-2 fitting
	Success	Noisy data	Approximate curve	Tests robustness
	Failure/Edge	Single data point	Cannot fit	Insufficient data
	Failure/Edge	Duplicate x-values issue	Poor fit/handled	Edge case
11	Power and Exponential Curve Fitting			
	Type	Input	Expected Output	Explanation
	Success	Power-law data	Power curve fitted	Validates transformation
	Success	Exponential data	Exponential curve fitted	Checks log conversion
	Success	Positive dataset	Parameters computed	Tests fitting
	Failure/Edge	Zero/negative y-values	Error	Log not defined
	Failure/Edge	Insufficient points	Cannot fit	Boundary case
12	Data Fitting and Visualization			
	Type	Input	Expected Output	Explanation
	Success	Real dataset	Best-fit model	Validates fitting
	Success	Dataset + plot	Curve visualized	Checks visualization
	Success	Error metric	Interpretation given	Tests analysis
	Failure/Edge	Missing values	Error/handled	Data cleaning needed
	Failure/Edge	Non-numeric data	Error	Invalid for fitting

CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	3	2	2	2	3	0	0	1	1	0	1	2	3
C02	3	2	2	2	3	0	0	1	1	0	1	2	3
C03	3	3	2	2	3	0	0	1	1	0	1	2	3
C04	3	2	2	2	3	0	0	1	1	0	1	2	3
C05	3	3	2	3	3	1	0	1	2	0	2	2	3

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
C01	Computing eigenvalues and solving nonlinear equations supports engineering mathematics knowledge (P01), problem analysis (P02), and solution development (P03). Numerical computation and visualization strongly support modern tool usage (P05) and experimental investigation (P04).	Supports development of computational mathematical solutions in software systems (PS01) and strong application of numerical techniques to real-world problems (PS02).
C02	Solving linear systems using iterative methods supports engineering knowledge (P01), problem analysis (P02), and investigation of convergence behavior (P04). Implementation using Python/MATLAB supports modern tool usage (P05).	Supports implementation of numerical algorithms in software systems (PS01) and application of computational methods to engineering problems (PS02).
C03	Applying numerical methods for differential equations supports engineering knowledge (P01), problem analysis (P02), and solution development (P03). Methods like Runge-Kutta involve investigation (P04) and tool usage (P05).	Supports development of simulation-based software solutions (PS01) and application of numerical techniques in real-world engineering systems (PS02).
C04	Performing curve fitting using least squares supports data analysis (P02), solution development (P03), and investigation of model fitting (P04). Computational tools support modern tool usage (P05).	Supports implementation of data modeling techniques in software systems (PS01) and application of statistical/numerical methods in practical problems (PS02).
C05	Analyzing and visualizing real datasets strongly supports problem analysis (P02), investigation (P04), and modern tool usage (P05). It also relates to environmental awareness (P06) and lifelong learning (P011), with communication of insights (P09).	Supports data-driven software development (PS01) and strong application of computational and visualization techniques to real-world problems (PS02).

DATABASEMANAGEMENTSYSTEMSLAB

Course	B.Tech.-III-Sem.	L	T	P	C
Course Code	25CSPC35	-	-	2	1

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
C01	create relational databases using SQL and constraints	4,9
C02	perform data operations and queries using SQL	4,9
C03	implement views, triggers, and procedures	4,9
C04	apply normalization and build database applications	4,9
C05	use MongoDB and integrate SQL with NoSQL systems	4,9

List of Experiments

Note: Take any database application and conduct experiments to get expertise on various case studies

Week	Title/Experiment
1	Creation of Tables using SQL: Data types, DDL commands (CREATE, ALTER, DROP) with primary and foreign keys.
2	DML operations: INSERT, SELECT, UPDATE, DELETE on tables.
3	SQL queries using operators: ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, EXCEPT, constraints.
4	Sub Queries: Nested and Correlated; Joins: Inner and Outer.
5	Aggregate functions (COUNT, SUM, AVG, MAX, MIN), GROUP BY, HAVING; creation and management of views.
6	Triggers: Creation and execution of INSERT, UPDATE, DELETE triggers.
7	Stored procedures: Create, execute, and modify stored procedures.
8	Application: Develop a mini database application using SQL.
9	Normalization: Convert given tables into 1NF, 2NF, and 3NF.
10	NoSQL database (MongoDB): Perform CRUD operations on collections.
11	NoSQL Database: Implement arrays, functions (count, sort, aggregate) using MongoDB.
12	Integration Task: Build a simple application integrating SQL and MongoDB operations.
References	

1. Database Management Systems Lab Manual, Department of CSE, CMRIT, Hyd.
Micro-Projects: Student should submit a report on <i>design, implement, test and debug</i> from one of the following/any other micro-project(s) approved by the faculty before the deadline.
1. Student Information Management System using SQL and Relational Database. 2. Online Book Store Database with Order and Payment Management. 3. Hospital Management Database System with Patient Records and Billing. 4. Inventory Control System with Supplier and Stock Management. 5. Employee Payroll Database with Triggers and Stored Procedures. 6. Railway Reservation System with Seat Allocation and Ticket Booking. 7. Online Food Ordering System with Menu, Orders, and Delivery Tracking. 8. Banking Database System with Transactions and Account Management. 9. Social Media Data Management using NoSQL (MongoDB). 10. E-Commerce Product Catalog using MongoDB with Aggregation Queries.

Test Cases

Week	Test Cases for Experiments			
1	Table Creation using SQL			
	Type	Input	Expected Output	Explanation
	Success	Create Student table	Table created	Valid table creation
	Success	Create FK relationship	Tables linked	Foreign key validation
	Success	Add new column	Column added	ALTER operation
	Failure/Edge	Duplicate primary key	Error displayed	Constraint validation
	Failure/Edge	Missing referenced table	Error displayed	FK dependency
2	DML Operations			
	Type	Input	Expected Output	Explanation
	Success	Insert student record	Record inserted	Create operation
	Success	Select student records	Records displayed	Read operation
	Success	Update marks	Record updated	Update operation
	Failure/Edge	NULL in NOT NULL field	Error displayed	Constraint validation
	Failure/Edge	Delete unmatched row	0 rows affected	Safe delete
3	SQL Operators and Set Operations			
	Type	Input	Expected Output	Explanation

	Success	Query using IN	Matching rows displayed	IN operator
	Success	Query using EXISTS	Valid rows displayed	EXISTS operator
	Success	UNION query	Combined rows shown	Set operation
	Failure/Edge	UNION mismatch columns	Error displayed	Compatibility check
	Failure/Edge	Empty subquery	Logical result shown	Edge condition
4	Subqueries and Joins			
	Type	Input	Expected Output	Explanation
	Success	Nested subquery	Correct rows displayed	Subquery execution
	Success	INNER JOIN query	Matching rows shown	Join validation
	Success	LEFT JOIN query	NULL rows included	Outer join
	Failure/Edge	Join without ON	Unexpected rows displayed	Logical mistake
	Failure/Edge	Incorrect subquery condition	Incorrect output shown	Weak logic
5	Aggregate Functions and Views			
	Type	Input	Expected Output	Explanation
	Success	COUNT with GROUP BY	Group counts displayed	Aggregation
	Success	HAVING clause query	Filtered groups shown	HAVING validation
	Success	Create VIEW	View created	View management
	Failure/Edge	Invalid GROUP BY query	Error displayed	Aggregation rule
	Failure/Edge	Drop base table	View invalid	Dependency issue
6	Triggers			
	Type	Input	Expected Output	Explanation
	Success	INSERT trigger	Log updated	Insert trigger
	Success	UPDATE trigger	Audit row created	Update trigger
	Success	DELETE trigger	Backup stored	Delete trigger
	Failure/Edge	Invalid trigger column	Error displayed	Schema validation
	Failure/Edge	Recursive trigger update	Recursion error displayed	Logical issue
7	Stored Procedures			
	Type	Input	Expected Output	Explanation
	Success	Procedure insert	Record inserted	Procedure execution
	Success	Average calculation procedure	Average displayed	Output procedure
	Success	Update procedure	Record updated	Procedure modification
	Failure/Edge	Missing parameter	Error displayed	Parameter

				validation
	Failure/Edge	Duplicate key insertion	Constraint error	Data integrity
8	Mini Database Application using SQL			
	Type	Input	Expected Output	Explanation
	Success	Add student data	Record saved	Application insert
	Success	Search student	Record displayed	Application query
	Success	Update student details	Record updated	Application update
	Failure/Edge	Invalid RollNo	No record found	Input validation
	Failure/Edge	SQL injection input	Query blocked	Security validation
9	Normalization			
	Type	Input	Expected Output	Explanation
	Success	Repeating attributes	Converted to 1NF	Removes redundancy
	Success	Partial dependency	Converted to 2NF	Dependency removal
	Success	Transitive dependency	Converted to 3NF	Normalization
	Failure/Edge	Already normalized table	No decomposition	Avoids redundancy
	Failure/Edge	Lossy decomposition	Rejected decomposition	Lossless design
10	MongoDB CRUD Operations			
	Type	Input	Expected Output	Explanation
	Success	insertOne() document	Document inserted	Create operation
	Success	find() query	Documents displayed	Read operation
	Success	updateOne() query	Document updated	Update operation
	Failure/Edge	deleteOne() unmatched	0 documents deleted	Safe delete
	Failure/Edge	Duplicate _id	Error displayed	Constraint handling
11	MongoDB Arrays, Count, Sort, Aggregate			
	Type	Input	Expected Output	Explanation
	Success	Array document	Array stored	Array handling
	Success	countDocuments() query	Count displayed	Count operation
	Success	sort() query	Sorted output shown	Sorting validation
	Failure/Edge	Empty collection aggregate	Empty result shown	Boundary case
	Failure/Edge	Missing sort field	Output generated	Schema-less handling
12	Sorting			
	Type	Input	Expected Output	Explanation
	Success	Store student data in SQL	Record saved	Relational storage

	Success	Store activity logs in MongoDB	Document inserted	NoSQL storage
	Success	Fetch combined student profile	Combined output displayed	Database integration
	Failure/Edge	Missing MongoDB record	Partial profile shown	Missing data handling
	Failure/Edge	Invalid student ID	Error displayed	Invalid retrieval

CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	3	2	3	1	3	0	0	1	1	0	1	3	2
C02	3	3	3	2	3	0	0	1	1	0	1	3	2
C03	3	3	3	2	3	0	0	1	1	0	1	3	2
C04	3	3	3	3	2	0	0	1	1	0	1	3	2
C05	3	3	3	3	3	0	0	1	1	0	1	3	3

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
C01	Creating relational databases using SQL and constraints supports engineering knowledge (P01), problem analysis (P02), and strong solution development (P03). Use of SQL tools strongly supports modern tool usage (P05).	Strongly supports database design and development in software systems (PS01) and application of relational database technologies (PS02).
C02	Performing data operations and queries strongly supports problem analysis (P02), solution development (P03), and investigation of query results (P04). SQL execution strongly supports tool usage	Strongly supports implementation of database operations in software systems (PS01) and application of query processing techniques (PS02).

	(P05).	
CO3	Implementing views, triggers, and procedures supports engineering knowledge (P01), problem analysis (P02), and solution development (P03). These involve investigation of database behavior (P04) and use of advanced DBMS tools (P05).	Strongly supports development of dynamic and automated database systems (PS01) and application of advanced database features (PS02).
CO4	Applying normalization and building database applications strongly supports problem analysis (P02), solution development (P03), and investigation of data redundancy and integrity (P04).	Strongly supports development of structured and efficient database systems (PS01) and application of design principles in real-world systems (PS02).
CO5	Using MongoDB and integrating SQL with NoSQL systems strongly supports problem analysis (P02), solution development (P03), and investigation of modern data systems (P04). Strong use of tools like MongoDB supports P05.	Strongly supports development of hybrid database-driven applications (PS01) and strong application of cutting-edge NoSQL technologies (PS02).

OOP THROUGH JAVA LAB

Course	B.Tech.-III-Sem.	L	T	P	C
Course Code	25CSPC36	-	-	2	1

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
CO1	develop Java programs using basic concepts	4,9
CO2	apply OOP concepts in Java	4,9
CO3	implement exception handling and multithreading	4,9
CO4	create GUI applications using Swing	4,9
CO5	build web and database applications using Java	4,9

List of Experiments

- Note:** 1. Use Eclipse or NetBeans platform and get acquainted with the various menus.
2. Faculty provides test cases for each experiment during labs for practice and validation.

Week	Title/Experiment
1	Write a Java program to a) find the roots of quadratic equation $ax^2+bx+c = 0$ b) print prime numbers up to a given integer (use BufferedReader)
2	Write a Java program to a) check whether a given string is a palindrome or not b) sort a list of strings using command-line input
3	Write a Java program to demonstrate a) method overloading and method overriding b) inheritance (single, multilevel) and interfaces (for multiple inheritance)
4	Write a Java program to demonstrate a) packages b) abstract classes and interfaces
5	Write a Java program to a) demonstrate exception handling b) create user-defined exception
6	Write a Java program to implement the producer-consumer problem using threads.
7	Write a Swings program to a) mouse and keyboard events b) factorial computation
8	Develop a swing application using JList, JTree, JTable, JTabbedPane and JScrollPane
9	Write a Java program for a simple calculator using GridLayout.
10	Write a servlet to a) handle parameter passing (employee login page) b) implement session tracking using cookies
11	Write a Java program to perform JDBC operations (connect, insert, update, delete)
12	Develop a mini web application using Servlets and JDBC
References	
1. OOP through JAVA Lab Manual, Department of CSE, CMRIT, Hyd.	
Micro-Projects: Student should submit a report on <i>design, implement, test and debug</i> from one of the following/any other micro-project(s) approved by the faculty before the deadline.	
1. Smart Student Record System using Encapsulation and File Handling. 2. Employee Payroll System using Inheritance and Polymorphism. 3. Library Management System using Classes and Objects. 4. Vehicle Rental System using Method Overriding and Dynamic Binding. 5. Banking System with Exception Handling and Abstraction. 6. Online Quiz Application using OOP Concepts and Collections Framework. 7. Hospital Management System using Packages and Access Modifiers. 8. Inventory Management System using Interfaces and Aggregation. 9. Online Student Portal using Servlets and JDBC. 10. E-Commerce Web Application using Servlets, JSP, and JDBC.	

Test Cases

Week	Test Cases for Experiments
1	Quadratic Roots & Prime Numbers
	a. Quadratic Equation Roots

	Type	Input	Expected Output	Explanation
	Success	a=1, b=-5, c=6	Roots = 3, 2	Real roots
	Success	a=1, b=2, c=1	Roots = -1, -1	Equal roots
	Success	a=1, b=0, c=-4	Roots = 2, -2	Positive and negative roots
	Failure/Edge e	a=1, b=1, c=1	Imaginary roots	Negative discriminant
	Failure/Edge e	a=0, b=2, c=3	Not quadratic	a cannot be zero
	b. Prime Numbers up to N using BufferedReader			
	Type	Input	Expected Output	Explanation
	Success	N=10	2 3 5 7	Normal case
	Success	N=20	2 3 5 7 11 13 17 19	Multiple primes
	Success	N=2	2	Smallest prime
	Failure/Edge	N=1	No prime numbers	Boundary case
	Failure/Edge	N=-5	Invalid input	Negative value
	2	Palindrome & String Sorting		
	a. String Palindrome			
	Type	Input	Expected Output	Explanation
	Success	madam	Palindrome	Normal palindrome
	Success	level	Palindrome	Odd-length palindrome
	Success	hello	Not Palindrome	Non-palindrome
	Failure/Edge	a	Palindrome	Single character
	Failure/Edge	Madam	Not Palindrome	Case-sensitive check
	b. Sort Strings using Command-line Input			
	Type	Input	Expected Output	Explanation
	Success	apple mango banana	apple banana mango	Alphabetical order
	Success	java python c	c java python	Different lengths
	Success	dog cat ant	ant cat dog	Normal sorting
	Failure/Edge	one	one	Single string
	Failure/Edge	Apple apple	Apple apple	Case-sensitive sorting
3	OOP Features			
	a. Method Overloading and Overriding			
	Type	Input	Expected Output	Explanation
	Success	add(10,5)	15	Overloaded integer method
	Success	add(2.5,3.5)	6.0	Overloaded double method
	Success	Child object calls overridden display()	Child display method executed	Overriding
	Failure/Edge e	Parent reference = Child object	Child method executed	Runtime polymorphism
	Failure/Edge e	Call undefined overloaded method	Compilation error	Invalid method signature
	b. Inheritance and Interfaces			
	Type	Input	Expected Output	Explanation

	Success	Single inheritance: Student extends Person	Parent and child methods accessible	Single inheritance
	Success	Multilevel: Animal → Dog → Puppy	All inherited methods accessible	Multilevel inheritance
	Success	Interface implementation for multiple inheritance	Interface methods executed	Multiple inheritance using interface
	Failure/Edge	Interface method not implemented	Compilation error	Interface rule
	Failure/Edge	Access private parent member directly	Compilation error	Encapsulation check
4	Packages & Abstract Classes			
	a. Packages			
	Type	Input	Expected Output	Explanation
	Success	Import user-defined package	Package class method executed	Package usage
	Success	Use built-in java.util.Scanner package	Input read successfully	Built-in package
	Success	Multiple classes in package	Correct class accessed	Package organization
	Failure/Edge	Missing import statement	Compilation error	Dependency error
	Failure/Edge	Wrong package name	Class not found error	Package path issue
	b. Abstract Classes and Interfaces			
	Type	Input	Expected Output	Explanation
	Success	Circle extends Shape	Area calculated	Abstract method implementation
	Success	Class implements Printable	Print method executed	Interface implementation
	Success	Abstract class with interface	Both methods executed	Combined use
	Failure/Edge	Instantiate abstract class	Compilation error	Abstract class cannot be instantiated
	Failure/Edge	Interface method not overridden	Compilation error	Interface contract
5	Exceptions			
	a. Exception Handling			
	Type	Input	Expected Output	Explanation
	Success	10 / 2	Result = 5	Normal division
	Success	10 / 0	ArithmeticException caught	Division by zero
	Success	Array index within range	Value displayed	Normal array access

	Failure/Edge	Invalid array index	ArrayIndexOutOfBoundsException caught	Boundary error
	Failure/Edge	Input abc for integer	NumberFormatException caught	Invalid input
	b. User-defined Exception			
	Type	Input	Expected Output	Explanation
	Success	Age=20	Valid age	Normal input
	Success	Marks=75	Valid marks	Valid range
	Success	Balance=5000	Valid balance	Normal condition
	Failure/Edge	Age=15	Custom exception: Invalid age	User-defined exception
	Failure/Edge	Marks=-10	Custom exception: Invalid marks	Range validation
6	Producer-Consumer Problem using Threads			
	Type	Input	Expected Output	Explanation
	Success	Produce item 1, consume item 1	Produced and consumed correctly	Basic synchronization
	Success	Produce items 1 to 5	All items consumed in order	Multiple operations
	Success	Buffer size = 1	Producer/consumer waits correctly	wait-notify validation
	Failure/Edge	Consumer starts before producer	Consumer waits, no crash	Empty buffer case
	Failure/Edge	No synchronization	Race condition may occur	Breaks incorrect logic
7	Swing Programs			
	a. Swing Mouse and Keyboard Events			
	Type	Input	Expected Output	Explanation
	Success	Mouse clicked on frame	Mouse clicked message displayed	Mouse event
	Success	Mouse moved	Coordinates displayed	Mouse motion event
	Success	Key pressed A	Key pressed message displayed	Keyboard event
	Failure/Edge	Press special key like Shift	Key event handled	Non-character key
	Failure/Edge	Multiple rapid key presses	Events handled correctly	Event handling robustness
	b. Swing Factorial Computation			
	Type	Input	Expected Output	Explanation
	Success	5	120	Normal factorial
	Success	0	1	Zero factorial
	Success	6	720	Standard case
	Failure/Edge	-3	Error message	Negative factorial invalid

	Failure/Edge	abc	Error message	Non-numeric input
8	Swing Components			
	Type	Input	Expected Output	Explanation
	Success	Select item in JList	Selected item displayed	JList validation
	Success	Expand node in JTree	Child nodes displayed	JTree validation
	Success	Show records in JTable with JScrollPane	Table displayed with scroll	JTable and scrollpane
	Failure/Edge	Empty list/table	Component shown without crash	Empty data
	Failure/Edge	Large table data	ScrollPane works correctly	UI scalability
9	Simple Calculator using GridLayout			
	Type	Input	Expected Output	Explanation
	Success	10 + 5	15	Addition
	Success	20 - 8	12	Subtraction
	Success	6 * 7	42	Multiplication
	Failure/Edge	10 / 0	Division by zero error	Invalid operation
	Failure/Edge	Operator pressed without number	Error message	UI input validation
10	Servlet (Parameters & Cookies)			
	a. Servlet Parameter Passing / Employee Login			
	Type	Input	Expected Output	Explanation
	Success	username=emp101, password=pass123	Login successful	Valid login
	Success	empId=101	Employee details displayed	Parameter handling
	Success	username and password submitted through form	Values received by servlet	Form processing
	Failure/Edge	Empty username/password	Error message	Missing input
	Failure/Edge	Wrong password	Login failed	Authentication check
	b. Session Tracking using Cookies			
	Type	Input	Expected Output	Explanation
	Success	Login creates cookie	Cookie stored in browser	Cookie creation
	Success	Visit next page with cookie	User session recognized	Session tracking
	Success	Logout clears cookie	Session ended	Cookie removal
	Failure/Edge	Cookies disabled	Session fallback needed	Cookie limitation
	Failure/Edge	Expired cookie	Login required again	Expiry handling

	e			
11	JDBC Operations			
	Type	Input	Expected Output	Explanation
	Success	Connect to database	Connection successful	JDBC connection
	Success	Insert employee record	Record inserted	Insert operation
	Success	Update employee salary	Record updated	Update operation
	Failure/Edge	Delete non-existing employee ID	0 rows affected	Safe delete
	Failure/Edge	Insert duplicate primary key	SQL exception	Constraint validation
12	Mini Web Application using Servlets and JDBC			
	Type	Input	Expected Output	Explanation
	Success	Add student/employee record	Data saved in database	Create operation
	Success	View all records	Records displayed on web page	Read operation
	Success	Update/delete record	Database updated	Full CRUD
	Failure/Edge	Invalid ID search	No record found	Input validation
	Failure/Edge	SQL injection input	Query blocked / safe error	Security validation

CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
CO1	3	2	2	1	3	0	0	1	1	0	1	3	2
CO2	3	3	3	2	3	0	0	1	1	0	1	3	2
CO3	3	3	3	2	3	0	0	1	1	0	1	3	2
CO4	3	2	3	2	3	0	0	1	2	0	1	3	3
CO5	3	3	3	3	3	0	0	1	2	1	1	3	3

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
CO1	Developing Java programs using basic concepts supports engineering knowledge (P01), basic problem analysis (P02), solution development (P03), and strong use of IDE/compiler tools (P05). Lab execution supports teamwork (P08) and communication of outputs (P09).	Strongly supports basic software development using Java (PS01) and application of programming tools for practical tasks (PS02).
CO2	Applying OOP concepts strongly supports problem analysis (P02), solution design (P03), and investigation of object-oriented behavior (P04). Use of	Strongly supports object-oriented software system development (PS01) and application of Java-based programming technologies (PS02).

	Java tools supports modern tool usage (P05).	
C03	Implementing exception handling and multithreading supports robust solution development (P03), problem analysis (P02), and investigation of concurrent program behavior (P04). Thread and exception handling practice supports modern tool usage (P05).	Strongly supports development of reliable and concurrent software systems (PS01) and application of modern Java programming techniques (PS02).
C04	Creating GUI applications using Swing supports solution development (P03), modern tool usage (P05), and user-interface communication (P09). Event handling involves analysis of application behavior (P02) and investigation (P04).	Strongly supports application development using GUI components (PS01) and application of modern interface development technologies (PS02).
C05	Building web and database applications using Java strongly supports engineering knowledge (P01), problem analysis (P02), solution development (P03), and investigation of application workflows (P04). JDBC and servlet implementation support tool usage (P05) and project awareness (P010).	Strongly supports development of database-driven web applications (PS01) and application of modern web/database technologies (PS02).

DATA EXPLORATION AND VISUALIZATION - PYTHON/ POWER BI

Course	B.Tech.-III-Sem.	L	T	P	C
Course Code	25CDSD31	-	-	2	1

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
C01	apply data cleaning and preprocessing techniques	4,9
C02	perform exploratory data analysis using Python	4,9
C03	analyze data using statistical and correlation methods	4,9
C04	create visualizations using Python libraries	4,9
C05	design interactive dashboards using Power BI	4,8,9

List of Experiments

Note: Faculty provides test cases for each experiment during labs for practice and validation.

Week	Title/Experiment
Data Exploration	
1	Setup Python environment and install required libraries.
2	Load datasets and perform preprocessing using Pandas (data types, indexing, filtering).
3	Perform data cleaning: handling missing values, duplicates, and outliers.
4	Generate descriptive statistics and summary reports.
5	Perform correlation and covariance analysis using numerical methods.
6	Perform EDA on a real-world dataset with visualization and interpretation.
Data Visualization	
7	Create basic plots using Matplotlib (line, bar, histogram).
8	Generate advanced visualizations using Seaborn (heatmaps, pairplots).
9	Build interactive visualizations using Plotly.
10	Perform time-series visualization and trend analysis.
11	Create dashboards using Power BI: data import, transformation, and basic visualizations.
12	Develop a dashboard using Power BI on a dataset.
References	
1. Data Exploration and Visualization Lab Manual, Dept. of CSE, CMRIT.	
Micro-Projects: Student should submit a report on <i>design, implement, test and debug</i> from one of the following/any other micro-project(s) approved by the faculty before the deadline.	
1. Analyze student performance using EDA techniques. 2. Explore COVID-19 dataset and extract insights. 3. Clean and analyze sales dataset for trends. 4. Perform employee attrition analysis using Python. 5. Detect anomalies in financial datasets. 6. Create interactive sales dashboard using Power BI. 7. Visualize population trends using Python plots. 8. Analyze stock market trends using time-series visualization. 9. Build social media analytics dashboard. 10. Create geographic data visualization using Power BI maps.	

Test Cases

Week	Test Cases for Experiments
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1	Setup Python Environment			
	Type	Input	Expected Output	Explanation
	Success	Install libraries	Libraries installed	Validates setup
	Success	Import libraries	No error	Checks environment
	Success	Run sample script	Output generated	Confirms readiness
	Failure/Edge	Missing library	Import error	Detects dependency issue
	Failure/Edge	Version conflict	Execution error	Checks compatibility
2	Load Dataset & Preprocessing			
	Type	Input	Expected Output	Explanation
	Success	Valid CSV	DataFrame loaded	Validates loading
	Success	Apply filtering	Filtered data	Checks indexing
	Success	Data types check	Correct types shown	Tests preprocessing
	Failure/Edge	Missing file	File error	Handles invalid path
	Failure/Edge	Corrupt dataset	Load error	Detects bad data
3	Data Cleaning			
	Type	Input	Expected Output	Explanation
	Success	Missing values	Values handled	Validates cleaning
	Success	Duplicate rows	Removed	Checks duplicates
	Success	Outliers present	Outliers detected	Tests detection
	Failure/Edge	All values missing	Error/handled	Boundary case
	Failure/Edge	Invalid data types	Cleaning fails	Detects issue
4	Descriptive Statistics			
	Type	Input	Expected Output	Explanation
	Success	Numeric dataset	Mean, median etc.	Validates statistics
	Success	Mixed data	Summary shown	Checks report
	Success	Large dataset	Stats computed	Tests scalability
	Failure/Edge	Non-numeric data	Partial stats	Edge case
	Failure/Edge	Empty dataset	Error	Boundary condition
5	Correlation & Covariance			
	Type	Input	Expected Output	Explanation
	Success	Numeric features	Correlation matrix	Validates relation
	Success	Positive relation	High correlation	Checks logic
	Success	Negative relation	Negative correlation	Tests inverse relation
	Failure/Edge	Constant column	NaN/undefined	Edge case
	Failure/Edge	Non-numeric data	Error	Validation check

6	EDA on Dataset			
	Type	Input	Expected Output	Explanation
	Success	Real dataset	Insights generated	Validates EDA
	Success	Visualization + stats	Patterns found	Checks analysis
	Success	Feature trends	Interpretation given	Tests reasoning
	Failure/Edge	Poor dataset	Weak insights	Edge case
	Failure/Edge	Missing columns	Error	Detects data issue
7	Matplotlib Basic Plots			
	Type	Input	Expected Output	Explanation
	Success	Numeric data	Line plot	Validates plotting
	Success	Category data	Bar chart	Checks representation
	Success	Distribution data	Histogram	Tests visualization
	Failure/Edge	Empty data	No plot	Boundary case
	Failure/Edge	Wrong axis input	Error	Detects mismatch

8	Seaborn Advanced Plots			
	Type	Input	Expected Output	Explanation
	Success	Numeric dataset	Heatmap generated	Validates correlation
	Success	Multi-variable data	Pairplot created	Checks relationships
	Success	Labeled data	Clear visualization	Tests readability
	Failure/Edge	Missing values	Plot issues	Edge case
	Failure/Edge	Invalid input	Error	Detects misuse
9	Plotly Interactive Visualization			
	Type	Input	Expected Output	Explanation
	Success	Dataset	Interactive chart	Validates Plotly
	Success	Hover action	Tooltip shown	Checks interactivity
	Success	Multiple traces	Combined plot	Tests features
	Failure/Edge	Invalid data format	Error	Detects issue
	Failure/Edge	Empty dataset	Blank chart	Boundary case
10	Time-Series Visualization			
	Type	Input	Expected Output	Explanation
	Success	Time-stamped data	Trend plot	Validates time-series
	Success	Seasonal data	Pattern shown	Checks trend
	Success	Large timeline	Smooth plot	Tests scalability
	Failure/Edge	Missing timestamps	Error	Detects missing index
	Failure/Edge	Irregular intervals	Distorted trend	Edge case
11	. Power BI Dashboard (Basic)			
	Type	Input	Expected Output	Explanation
	Success	Import dataset	Data loaded	Validates import
	Success	Create visuals	Charts displayed	Checks dashboard
	Success	Apply filters	Updated visuals	Tests interactivity
	Failure/Edge	Invalid data source	Load error	Detects issue
	Failure/Edge	Missing columns	Visualization error	Checks data mapping
12	Power BI Dashboard (Complete)			
	Type	Input	Expected Output	Explanation
	Success	Full dataset	Dashboard created	Validates design
	Success	Multiple visuals	Integrated report	Checks completeness
	Success	User interaction	Dynamic updates	Tests usability
	Failure/Edge	Poor data model	Incorrect visuals	Detects design flaw
	Failure/Edge	Large dataset	Performance issue	Edge case

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CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PS01	PS02
C01	3	3	2	2	3	0	0	1	1	0	1	3	3
C02	3	3	2	3	3	0	0	1	1	0	1	3	3
C03	3	3	2	3	3	0	0	1	1	0	1	3	3
C04	3	2	2	2	3	0	0	1	2	0	1	3	3
C05	3	2	3	2	3	1	0	1	2	1	1	3	3

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
C01	Applying data cleaning and preprocessing techniques supports engineering knowledge (PO1), strong problem analysis (PO2), and investigation of data quality issues (PO4). Use of Python libraries strongly supports modern tool usage (PO5).	Strongly supports development of data processing pipelines in software systems (PS01) and application of data engineering techniques using modern tools (PS02).
C02	Performing exploratory data analysis strongly supports problem analysis (PO2), investigation of datasets (PO4), and engineering knowledge (PO1). Visualization and analysis using Python support modern tool usage (PO5).	Strongly supports implementation of data analysis modules in software systems (PS01) and application of data analytics technologies (PS02).
C03	Analyzing data using statistical and correlation methods strongly supports engineering knowledge (PO1), problem analysis (PO2), and investigation of relationships in data (PO4).	Strongly supports data-driven decision-making in software systems (PS01) and application of statistical techniques in real-world problems (PS02).

	Python-based computation supports modern tool usage (P05).	
C04	Creating visualizations using Python libraries supports solution development (P03), communication of insights (P09), and strong use of visualization tools (P05).	Strongly supports development of visualization components in software systems (PS01) and application of modern visualization technologies (PS02).
C05	Designing interactive dashboards using Power BI supports solution development (P03), communication (P09), and modern tool usage (P05). It also relates to societal/business applications (P06) and basic project/reporting skills (P010).	Strongly supports development of dashboard-based software solutions (PS01) and application of cutting-edge BI tools for real-world data problems (PS02).